

TEST STRATEGY

DemoQA Book Store Application — Login Module

Field	Value
Application	DemoQA Book Store Application
Module	Login (https://demoqa.com/login)
Document ID	TS-DEMOQA-LOGIN-001
Version	1.0
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1. Purpose

This Test Strategy defines the overall approach for testing the Login module of the DemoQA Book Store Application. It establishes the testing techniques, levels, risk priorities, environment, and entry/exit criteria that all Login test activities will follow. Detailed test cases, schedules, and staffing are covered in the accompanying Test Plan.

2. Scope

In Scope

- Login UI: username field, password field, login button, field validation
- Authentication: valid and invalid credential combinations
- Error handling and validation messages, password masking
- Session behavior: login, logout, unauthorized/direct URL access, back/refresh after logout
- Boundary, special-character, and negative input handling
- Browser compatibility (Chrome, Firefox, Edge) and baseline performance

Out of Scope

- New user registration, user profile, book search/management/purchase
- Full backend/database testing and formal penetration testing

3. Test Approach

Testing follows a risk-based, requirement-driven approach. Requirements are analyzed for functional, validation, security, UI, and session behavior, then translated into test conditions using the techniques below.

Technique	Application to Login Module
Equivalence Partitioning	Group valid vs. invalid username/password inputs
Boundary Value Analysis	Test field length limits and edge-case input sizes
Decision Table Testing	Cover username/password valid-invalid-empty combinations
State Transition Testing	Verify Logged Out → Logged In → Logged Out flow
Use Case Testing	Validate the complete end-to-end login journey

Technique	Application to Login Module
Error Guessing	Target likely failure points (e.g., SQL/special characters)
Exploratory Testing	Investigate unexpected or undocumented behavior

Execution proceeds in stages: Smoke Testing (build acceptance) → Functional & Negative Testing → Security/Session Testing → UI & Compatibility Testing → Defect Reporting → Retesting → Regression Testing → Test Summary Reporting.

4. Test Levels and Types

Type	Focus
Functional	Valid and invalid authentication scenarios
Validation	Required-field and input-format checks
Negative	Invalid, empty, and boundary/special-character inputs
Security	Unauthorized access, credential handling, password masking
Session	Login/logout, session expiry, back-navigation after logout
UI/Usability	Field alignment, labels, buttons, readability
Compatibility	Chrome, Firefox, Edge (desktop; mobile where applicable)
Performance	Baseline login response time under normal conditions

5. Risk-Based Test Priorities

Risk	Impact	Testing Response
Valid user cannot log in	Critical	High-priority functional testing
Invalid user gains access	Critical	Security/negative testing
Password shown in plain text	High	UI/security testing
Logout does not end session	High	Session testing
Incorrect error messages	Medium	Validation testing
Login is slow	Medium	Performance testing
Login fails on a browser	Medium	Compatibility testing

6. Test Environment

- Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge
- OS: Windows; Android/mobile where applicable
- Network: normal and slow/intermittent conditions
- Tools: JIRA/Zoho Sprints (defects), Excel (scenarios/cases), Browser DevTools, screen-recording tool
- Configuration recorded per run: build/version, OS, browser, device, resolution, network, test data, date

7. Entry, Suspension, and Exit Criteria

Entry Criteria

- Login page is accessible and the build is stable enough for smoke testing

- Test cases, test data, and environment are ready

Suspension Criteria

- Login page/button is unavailable or the application crashes
- Authentication service is down or valid test credentials are missing

Exit (Completion) Criteria

- All planned tests executed; critical login functionality passes
- Critical/High defects fixed or formally accepted; retesting and regression complete
- Test Summary Report reviewed and approved by QA Lead/stakeholder

8. Defect Management

Defects follow the lifecycle New → Assigned → Open → Fixed → Retest → Verified → Closed (or Retest Failed → Reopened on failure). Each defect is logged with severity (Critical/High/Medium/Low) and priority (P1–P3); for example, login accepting an incorrect password is classified Critical severity, P1 priority. Every fix is retested, followed by regression testing of related login functionality (valid/invalid login, empty fields, logout, session behavior).

9. Deliverables

Deliverable	Purpose
Test Plan	Defines detailed scope, resources, and schedule
Test Scenarios & Cases	Identify and detail functionality to test
Bug Reports	Document defects found
Execution & Retest Reports	Record results and fix verification
Test Summary Report	Final testing status and recommendation

10. Roles and Responsibilities

Role	Responsibility
QA/Test Engineer	Design, execute, report, retest, and regression test
QA Lead	Review test artifacts and monitor quality
Developer	Analyze and fix reported defects
Project Manager	Manage schedule and resources
Business Stakeholder	Clarify requirements and accept results